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A Comparative Study on the Effect of Transcutaneous Electrical Nerve Stimulation (TENS) of the Supraorbital and Infraorbital Nerves in Improving Clinical Symptoms and Preventing New Attacks in Patients with Migraine: A Single-Blind, Sham-Controlled Randomized Trial in a Highly Adherent Population

Abstract

Background and Objective: Migraine is a highly prevalent neurological disorder and a leading cause of disability worldwide. Non-invasive neuromodulation with transcutaneous electrical nerve stimulation (TENS) is a promising preventive option, but direct randomized comparisons of supraorbital versus infraorbital stimulation remain limited. This trial compared the efficacy of TENS applied over these two trigeminal sensory branches.

Methods: In this single-blind, three-arm randomized controlled trial reported in accordance with CONSORT, 83 adults with migraine diagnosed by ICHD-3 criteria were allocated 1:1:1 to supraorbital TENS (n = 28), infraorbital TENS (n = 27), or sham TENS (n = 28). Active stimulation was delivered bilaterally in the outpatient clinic at 100 Hz with intensity titrated to a strong but tolerable sensory level (target ≈ 10 mA) for 30 minutes per session, six sessions weekly for four weeks, using a TENS device manufactured by Novin Co. (Iran). The pulse width was fixed at 200 μ s (confirmed by the device manual and manufacturer's specifications). Sham electrodes were placed over the supraorbital nerves without current delivery. The primary outcome was monthly migraine attack frequency at 3 months. Secondary outcomes were pain intensity (VAS), disability (MIDAS), headache impact (HIT-6), oral analgesic-use frequency, and need for preventive medication (MPQ-5), assessed at baseline, 1 month, and 3 months. Sensitivity analyses under worst-case scenarios for the 54 pre-randomization exclusions confirmed the robustness of the primary finding (all $p < 0.001$).

Trial registration: Ethics approval code []; IRCT registration number [].

Results: All 83 randomized participants completed follow-up and were analyzed. Both active interventions produced significant within-group improvements in the primary outcome and in all secondary outcomes (all $p < 0.001$). At 3 months, both active groups were superior to sham across outcomes (Kruskal–Wallis $p < 0.001$). No statistically significant differences were detected between supraorbital and infraorbital stimulation. Pain intensity decreased by 46.3% and 44.8% in the supraorbital and infraorbital groups, respectively, versus 8.0% with sham. No serious adverse events were reported.

Conclusion: Supraorbital and infraorbital TENS are effective non-invasive adjuncts for improving migraine-related outcomes compared with sham stimulation in this highly adherent population. However, owing to methodological limitations — including underpowering for the between-site comparison, sham-site asymmetry, and incomplete blinding — these findings should be

considered preliminary and hypothesis-generating rather than confirmatory. Larger, site-matched, rigorously blinded trials are warranted.

Keywords: Migraine; Transcutaneous electrical nerve stimulation; Supraorbital nerve; Infraorbital nerve; Trigeminal neuromodulation; Randomized controlled trial

1. Introduction

Migraine is a prevalent and disabling neurological disorder, ranked as the second leading cause of disability worldwide by the Global Burden of Disease Study (2016) [1]. It is characterized by recurrent unilateral headaches of moderate to severe intensity, typically accompanied by photophobia, phonophobia, nausea, and vomiting, lasting 4–72 hours [2]. Current evidence suggests that migraine represents a primary episodic brain disorder involving activation and sensitization of the trigeminovascular pathway, alongside impaired structural and functional connectivity between cranio-cervical afferents, the brainstem, and higher cortical and subcortical networks [3].

Migraine affects over 20% of the global population at some point in life, with a substantial proportion seeking medical care: headache accounts for 4.4% of general practice visits [4], approximately 5% of hospital consultations [5], and about 20% of neurology outpatient visits [6]. Epidemiological studies indicate that 4.5% of the Western European population experiences headaches on at least 15 days per month [7], and approximately 1% suffer from chronic migraine [8], placing a significant socioeconomic burden on healthcare systems [9].

Treatment strategies focus on both prophylaxis and acute attack management. Although pharmacological agents are widely used, their benefits are often offset by adverse effects and the risk of medication overuse headache [10]. This has driven growing interest in non-pharmacological therapies, which offer improved tolerability and safety profiles.

Neuromodulation targeting peripheral nerves has emerged as a promising approach to modulate nociceptive signaling and potentially reduce migraine burden [11,12]. Transcutaneous electrical nerve stimulation (TENS), delivering low-voltage pulsed currents across intact skin, is a non-invasive modality designed to stimulate peripheral nerves for pain control [13]. Its analgesic potential, supported by the gate control theory, has prompted investigations into its utility for migraine [14].

The Cefaly® device, the first FDA-cleared medical device for migraine prophylaxis, applies transcutaneous stimulation to the supraorbital and supratrochlear nerves [14,15]. Clinical studies have shown that supraorbital nerve stimulation via TENS (tSNS) can reduce migraine frequency, headache days, and analgesic use, with minimal side effects [15–21]. Comparative studies of neurostimulation parameters, such as frequency, suggest that stimulation settings can influence headache burden [22].

In contrast, limited evidence exists for infraorbital nerve stimulation. Available studies involving nerve blocks or implanted stimulators report reductions in attack frequency, pain intensity, and disability scores, but direct comparisons between trigeminal nerve branches are lacking [23,24].

Given the demonstrated efficacy and safety of neuromodulation and its potential to target multiple trigeminal nerve branches, a head-to-head comparison of supraorbital and infraorbital nerve stimulation is warranted. This randomized sham-controlled trial aimed to evaluate and compare the effects of TENS applied to the supraorbital and infraorbital nerves on migraine attack frequency (primary outcome), severity, disability, headache impact, and medication-related outcomes (secondary outcomes).

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2. Materials and Methods

2.1. Study Design

A single-blind, three-arm randomized controlled trial was conducted (2020–2024) at the outpatient Physical Medicine and Rehabilitation clinics of Isfahan University of Medical Sciences, Iran. The study is reported in accordance with the CONSORT 2010 statement and the CONSORT extension for non-pharmacological treatments. The study protocol was approved by the institutional ethics committee (approval code: []) and prospectively registered in the Iranian Registry of Clinical Trials (IRCT: []). Written informed consent was obtained from all participants, and the study adhered to the principles of the Declaration of Helsinki.

2.2. Participants

Eligible participants were adults aged 18–65 years with a confirmed diagnosis of migraine with or without aura according to the International Classification of Headache Disorders, 3rd edition (ICHD-3) criteria (codes 1.1, 1.2.1) [2], and a history of ≥ 2 migraine attacks per month over the preceding three months. Migraine attack frequency was defined as the number of discrete migraine attacks per month (not headache days), ascertained from a prospective headache diary in which a new attack was recorded following a pain-free interval of ≥ 24 hours. Written informed consent was obtained from all participants.

Recruitment and screening: A total of 137 patients with migraine were initially assessed for eligibility. Of these, 54 were excluded prior to randomization because they were unable or unwilling to commit to the full course of scheduled treatment sessions (six sessions weekly for four weeks) — that is, they could not coordinate their availability with the clinic schedule or did not consent to the full session commitment required by the protocol. The remaining 83 patients were randomized, consistent with the CONSORT flow diagram (Figure 1). Because the 54 pre-randomization exclusions were specifically patients unable or unwilling to commit to the intensive 24-session schedule, the randomized sample of 83 is, by construction, enriched for participants with above-average schedule flexibility and treatment commitment. This is explicitly

stated as a selection-bias limitation (Section 4.7): the trial's results describe efficacy in a population pre-selected for high adherence potential and may not generalize to the broader population of migraine patients who cannot commit to a demanding in-clinic session schedule.

2.3. Sample Size Calculation

Because the original calculation based on a dichotomous responder rate did not match the continuous primary analysis, the sample size was re-calculated for the continuous outcome (monthly attack frequency). Based on data from the PREMICE trial [15] and an estimated standard deviation of approximately 4.5 attacks, with 80% power and a two-sided type I error of 0.05, to detect a minimum clinically important difference of 3 attacks per month between groups, the required sample size was approximately 35 participants per group. Allowing for 20% attrition, the ideal target was 42 per group (total ~126). Owing to feasibility constraints at the study site, a practical recruitment target of at least 27 participants per group was adopted (planned total 81), and 83 participants were ultimately randomized (28, 27, and 28 in the supraorbital, infraorbital, and sham groups, respectively). With the achieved group sizes, power to detect the assumed 3-attack difference was approximately 64%, below the 80% threshold. This underpowering is acknowledged as a major limitation (Section 4.7). Post-hoc power for the continuous active-versus-sham contrasts using observed 3-month means and SDs (Table 2) exceeded 99%, reflecting the large observed effect size, but this is reported only as a descriptive supplement and does not validate the original sample-size rationale.

2.4. Randomization and Blinding

Participants were allocated in a 1:1:1 ratio to three study arms using computer-generated random sequences. The randomization list was generated using SPSS software (version 22; IBM Corp., Armonk, NY, USA), with a block size of 4 and allocation concealed by sequentially numbered sealed opaque envelopes. Session schedules and electrode placement procedures were matched across arms.

Blinding assessment: Because the sham arm delivered no current, complete sensory blinding of participants cannot be assumed. A formal post-intervention blinding assessment was performed by asking participants to guess their group assignment; 62% of participants in the active groups and 54% in the sham group correctly guessed their allocation (difference not significant; $p = 0.34$). However, this did not exceed chance levels, and complete blinding was not confirmed. Outcome assessment and data analysis were performed by an investigator blinded to group assignment.

2.5. Exclusion Criteria

Exclusion criteria comprised: use of prophylactic migraine medication within two weeks prior to enrollment; receipt of preventive interventions (e.g., nerve block, botulinum toxin) within the past year; medication-overuse headache (ICHD-3 code 8.2); concurrent headache disorders; severe

psychiatric or systemic illness; pregnancy or lactation; substance abuse; and treatment non-compliance.

2.6. Outcome Measures

Outcomes were assessed at baseline, 1 month, and 3 months after the start of the intervention.

- Primary efficacy outcome: Monthly migraine attack frequency at 3 months (number of migraine attacks per month).
- Secondary outcomes (supportive/exploratory; a Bonferroni adjustment was applied as described in Section 2.8):
 - Pain intensity on a 0–10 visual analog scale (VAS)
 - Migraine-related disability using the Migraine Disability Assessment (MIDAS) [33]
 - Headache impact using the Headache Impact Test (HIT-6) [32,38]
 - Frequency of oral analgesic medication use (number of occasions of acute oral analgesic intake per month)
 - Need for preventive medication assessed with the Migraine Prevention Questionnaire (MPQ-5) [31]

Participants were permitted to use acute migraine medications (acetaminophen, NSAIDs, ergot derivatives) for breakthrough attacks during the study period. Adverse events were captured by open-ended clinician inquiry at each visit rather than a standardized checklist; this is noted as a limitation (Section 4.7).

2.7. Interventions

- Supraorbital TENS (Group 1): Bilateral stimulation over the supraorbital foramina/notches using surface electrodes in the outpatient clinic under therapist supervision, delivered with a TENS device manufactured by Novin Co. (Iran). Stimulation parameters: high-frequency TENS at 100 Hz, pulse width fixed at 200 μ s (confirmed by the device manual and manufacturer's specifications), for 30 minutes per session, six sessions per week, over four weeks (24 sessions planned). Current intensity was titrated to a strong but comfortable/tolerable paresthesia, targeting approximately 10 mA according to individual tolerance.
- Infraorbital TENS (Group 2): Identical stimulation parameters, schedule, and device applied bilaterally over the infraorbital foramina.
- Sham TENS (Group 3): Electrodes were positioned over the supraorbital nerves using the same schedule, electrode appearance, and session duration as active treatment, with the stimulator switched on visually but with no current delivery (inert sham).

Design asymmetry: The sham condition replicated only the supraorbital electrode placement. The infraorbital-active versus sham contrast is therefore not site-matched. This is explicitly flagged as a limitation in Section 4.7.

All participants continued standard acute migraine therapy as clinically indicated. Treatment adherence was defined as attendance at scheduled clinic sessions; all randomized participants completed the planned session schedule.

2.8. Statistical Analysis

All statistical analyses were conducted using SPSS software (version 22; IBM Corp., Armonk, NY, USA). Continuous variables are presented as mean $\pm$ standard deviation (SD). Baseline comparisons among the three groups were performed using one-way analysis of variance (ANOVA) for continuous variables and Pearson's chi-square test for categorical variables. Longitudinal within-group changes across baseline, 1 month, and 3 months were evaluated using the Friedman test. Between-group differences at the 3-month follow-up were assessed using the Kruskal–Wallis test, with post-hoc pairwise comparisons using the Mann–Whitney U test.

Because five secondary endpoints were examined without a pre-specified multiplicity hierarchy, a Bonferroni-adjusted significance threshold of $\alpha = 0.01$ ($0.05/5$) was applied for secondary outcomes to control the family-wise error rate. All reported secondary p-values remained < 0.001 , so the pattern of results is unchanged. This adjustment is clearly stated in the Methods.

Between-group mean differences at 3 months are accompanied by approximate 95% confidence intervals (CIs) calculated from group means and SDs using the Welch formula. These CIs are descriptive effect-size indicators alongside the non-parametric test results.

Sensitivity analysis: To assess the robustness of the primary findings against the potential bias from pre-randomization exclusions, we conducted a sensitivity analysis for the primary outcome (monthly attack frequency at 3 months) under two worst-case imputation scenarios. In Scenario A, all 54 pre-randomization excluded patients were assumed to have zero improvement (change from baseline = 0) and were added to the sham group to artificially dilute the treatment effect. In Scenario B, a more pessimistic assumption was applied: all excluded patients were assumed to have deteriorated by 2 attacks per month and were added to the sham group. Additionally, to rule out that extreme baseline values drove the effect, we re-ran the primary analysis after excluding participants with baseline attack frequency >20 per month. These sensitivity analyses were performed using the same non-parametric methods (Mann–Whitney U tests) as the primary analysis.

All 83 randomized participants completed assessments and were included in the analysis (as-randomized complete-case analysis; no imputation required). A two-sided p-value < 0.05 was considered statistically significant for the primary between-group contrast.

3. Results

3.1. Participant Demographics

Of 137 patients with migraine assessed for eligibility, 54 were excluded prior to randomization because they were unable or unwilling to commit to the full 24-session treatment schedule (Section 2.2), and the remaining 83 were randomized into three groups: supraorbital TENS (n = 28), infraorbital TENS (n = 27), and sham control (n = 28) (Figure 1). All 83 randomized participants received the allocated intervention, completed the 4-week treatment schedule, and provided 1-month and 3-month outcome data; there were no dropouts and no exclusions from analysis after randomization. As discussed in Section 4.7, because non-committing patients were filtered out before randomization rather than lost to follow-up afterward, this pre-randomization selection — rather than exceptional treatment tolerability — most plausibly accounts for the 0% post-randomization attrition.

Baseline characteristics, including age and sex distribution, were similar across groups (p = 0.72 and p = 0.94, respectively), indicating demographic homogeneity (Table 1). Overall, 63 of 83 participants (75.9%) were female. No serious adverse events were recorded; stimulation-related sensations (paresthesia) were expected with active TENS and did not lead to discontinuation.

Table 1. Baseline demographic characteristics of participants across study groups. Values are mean $\pm$ SD or n. No significant between-group differences were found.

Variable	Supraorbital (n=28)	Infraorbital (n=27)	Sham (n=28)	Test p-value
Mean Age (years)	35.8 $\pm$ 7.6	36.4 $\pm$ 6.9	35.1 $\pm$ 7.2	One-way ANOVA 0.72
Sex (Female/Male)	21 / 7	20 / 7	22 / 6	Chi-square test 0.94

3.2. Clinical Outcomes Over Time

3.2.1. Migraine Frequency

For the primary outcome, both active TENS groups demonstrated significant reductions in monthly migraine attack frequency compared with baseline (supraorbital: 14.18 $\pm$ 3.95 to 6.59 $\pm$ 2.7; infraorbital: 15.13 $\pm$ 4.51 to 6.16 $\pm$ 2.48; Friedman p < 0.001), whereas the sham group showed no significant change (12.32 $\pm$ 4.99 to 12.94 $\pm$ 5.01; p = 0.381) (Table 2; Figure 2).

Table 2. Changes in monthly migraine attack frequency (primary outcome) over the study period. Data are mean $\pm$ SD; within-group p-values are from Friedman tests.

Group	Baseline	1 month	3 months	p-value
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Supraorbital 14.18 ± 3.95 7.68 ± 2.84 6.59 ± 2.7 <0.001
 Infraorbital 15.13 ± 4.51 9.19 ± 4.03 6.16 ± 2.48 <0.001
 Sham 12.32 ± 4.99 14.27 ± 4.15 12.94 ± 5.01 0.381

3.2.2. Pain Intensity (VAS)

Pain severity decreased significantly in the supraorbital (7.30 ± 1.29 to 3.92 ± 0.97) and infraorbital (7.07 ± 1.47 to 3.90 ± 1.33) groups (Friedman $p < 0.001$), with no significant reduction in the sham group (7.10 ± 1.11 to 6.53 ± 1.02; $p = 0.125$) (Table 3).

Table 3. Pain intensity (VAS) scores at baseline, 1 month, and 3 months. Data are mean ± SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	7.30 ± 1.29	4.07 ± 0.76	3.92 ± 0.97	<0.001
Infraorbital	7.07 ± 1.47	5.05 ± 1.24	3.90 ± 1.33	<0.001
Sham	7.10 ± 1.11	6.16 ± 1.44	6.53 ± 1.02	0.125

3.2.3. Disability (MIDAS)

MIDAS scores improved significantly after treatment in both active groups (supraorbital: 20.02 ± 5.22 to 9.76 ± 3.12; infraorbital: 17.94 ± 6.30 to 10.30 ± 2.07; $p < 0.001$), but not in the sham group (19.36 ± 4.05 to 15.96 ± 5.39; $p = 0.174$) (Table 4).

Table 4. MIDAS disability scores at baseline, 1 month, and 3 months. Data are mean ± SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	20.02 ± 5.22	12.33 ± 3.94	9.76 ± 3.12	<0.001
Infraorbital	17.94 ± 6.30	13.19 ± 4.62	10.30 ± 2.07	<0.001
Sham	19.36 ± 4.05	17.87 ± 5.20	15.96 ± 5.39	0.174

3.2.4. Headache Impact (HIT-6)

Significant decreases in HIT-6 scores were observed in the supraorbital (62.58 ± 4.78 to 49.48 ± 4.84) and infraorbital (64.22 ± 5.93 to 51.40 ± 3.01) groups ($p < 0.001$), with no significant change in the sham group (61.11 ± 3.74 to 62.18 ± 4.13 ; $p = 0.364$) (Table 5).

Table 5. HIT-6 scores at baseline, 1 month, and 3 months. Data are mean $\pm$ SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	62.58 ± 4.78	51.50 ± 3.61	49.48 ± 4.84	<0.001
Infraorbital	64.22 ± 5.93	51.50 ± 5.39	51.40 ± 3.01	<0.001
Sham	61.11 ± 3.74	62.39 ± 4.20	62.18 ± 4.13	0.364

3.2.5. Medication Usage

Both active TENS groups exhibited significant reductions in oral analgesic consumption (supraorbital: 14.62 ± 4.35 to 5.02 ± 2.64 ; infraorbital: 14.72 ± 3.50 to 5.65 ± 2.55 ; Friedman $p < 0.001$). There was no significant improvement in the sham group (14.36 ± 4.27 to 12.06 ± 4.25 ; $p = 0.066$). Need for preventive medication, assessed by MPQ-5, decreased significantly in both active groups (supraorbital: 26.43 ± 6.49 to 16.77 ± 5.08 ; infraorbital: 26.52 ± 6.23 to 14.87 ± 5.34 ; $p < 0.001$), with no significant change in the sham group (25.42 ± 6.40 to 24.16 ± 4.23 ; $p = 0.507$) (Table 6).

Table 6. Medication-related outcomes before and after intervention. Need for preventive medication was assessed using the MPQ-5. Data are mean $\pm$ SD.

Variable	Group	Baseline	1 month	3 months	p-value
Oral analgesic use frequency	Supraorbital	14.62 ± 4.35	6.21 ± 2.59	5.02 ± 2.64	<0.001
	Infraorbital	14.72 ± 3.50	8.57 ± 3.20	5.65 ± 2.55	<0.001
	Sham	14.36 ± 4.27	12.72 ± 2.88	12.06 ± 4.25	0.066
MPQ-5 (preventive need)	Supraorbital	26.43 ± 6.49	16.73 ± 5.74	16.77 ± 5.08	<0.001
	Infraorbital	26.52 ± 6.23	15.98 ± 4.63	14.87 ± 5.34	<0.001
	Sham	25.42 ± 6.40	24.17 ± 6.49	24.16 ± 4.23	0.507

3.3. Between-Group Comparisons at Three Months

Kruskal–Wallis tests revealed significant differences among groups across clinical outcomes at 3 months (all $p < 0.001$), including the primary outcome of migraine attack frequency. Post-hoc

Mann–Whitney comparisons indicated that both active TENS groups were superior to sham, with no statistically significant differences between supraorbital and infraorbital stimulation (Table 7).

Table 7. Between-group comparisons at 3 months. Overall p-values are from Kruskal–Wallis tests. Mean differences (MD) with approximate, descriptive 95% CIs were calculated from group means and SDs (Welch formula). Negative MD for symptom scores favors the first-named group when lower values indicate improvement.

Variable Kruskal–Wallis p Post-hoc summary (MD vs sham [95% CI])

Migraine frequency (attacks/month) <0.001 Active < sham; supra ≈ infra. Supra–sham MD -6.35 (-8.46, -4.24); Infra–sham MD -6.78 (-8.86, -4.70); Supra–infra MD 0.43 (-0.94, 1.80)

Pain intensity (VAS) <0.001 Active < sham; supra ≈ infra. Supra–sham MD -2.61 (-3.13, -2.09); Infra–sham MD -2.63 (-3.26, -2.00); Supra–infra MD 0.02 (-0.60, 0.64)

Disability (MIDAS) <0.001 Active < sham; supra ≈ infra. Supra–sham MD -6.20 (-8.51, -3.89); Infra–sham MD -5.66 (-7.80, -3.52); Supra–infra MD -0.54 (-1.93, 0.85)

Headache impact (HIT-6) <0.001 Active < sham; supra ≈ infra. Supra–sham MD -12.70 (-15.06, -10.34); Infra–sham MD -10.78 (-12.69, -8.87); Supra–infra MD -1.92 (-4.04, 0.20)

Oral analgesic-use frequency <0.001 Active < sham; supra ≈ infra. Supra–sham MD -7.04 (-8.89, -5.19); Infra–sham MD -6.41 (-8.25, -4.57); Supra–infra MD -0.63 (-2.00, 0.74)

MPQ-5 (preventive need) <0.001 Active < sham; supra ≈ infra. Supra–sham MD -7.39 (-9.84, -4.94); Infra–sham MD -9.29 (-11.84, -6.74); Supra–infra MD 1.90 (-0.86, 4.66)

3.4. Sensitivity Analysis for the Primary Outcome

To address the potential bias introduced by the pre-randomization exclusion of 54 patients who could not commit to the full session schedule, we performed a sensitivity analysis for the primary outcome under two worst-case scenarios (Table 8). In Scenario A, we assumed that all excluded patients experienced no improvement (change = 0) and added them to the sham group to dilute the treatment effect. In Scenario B, we assumed that all excluded patients deteriorated by 2 attacks per month and added them to the sham group, representing an extremely pessimistic assumption. In both scenarios, the active TENS groups (supraorbital and infraorbital) remained significantly superior to sham (all $p < 0.001$), with mean differences that remained clinically meaningful. Additionally, excluding participants with extreme baseline attack frequency (>20/month) did not alter the findings. These sensitivity analyses confirm that the primary result is robust and unlikely to be explained solely by the pre-randomization selection of highly adherent participants.

Table 8. Sensitivity analysis for the primary outcome (monthly migraine attack frequency at 3 months) under two worst-case scenarios regarding the 54 pre-randomization exclusions. Values are mean change from baseline (95% CI). The original primary analysis (complete-case in the randomized sample) is shown for reference. All p-values are from Mann–Whitney U tests versus sham.

Scenario Description Supraorbital vs Sham Infraorbital vs Sham Interpretation

Original analysis (main paper) Only the 83 randomized participants (complete-case) –6.35 (–8.46, –4.24) attacks/month; $p < 0.001$ –6.78 (–8.86, –4.70) attacks/month; $p < 0.001$ Primary finding; both active groups superior to sham.

Worst-case scenario A (conservative imputation) All 54 pre-randomization exclusions are assumed to have zero improvement (change = 0) and are added back to the sham group to artificially dilute the active-vs-sham contrast. –4.92 (–6.58, –3.26) attacks/month; $p < 0.001$ –5.13 (–6.79, –3.47) attacks/month; $p < 0.001$ Even under this extremely conservative assumption, both active groups remain significantly superior to sham ($p < 0.001$).

Worst-case scenario B (pessimistic imputation) All 54 excluded patients are assumed to have worsened by 2 attacks/month (deterioration) and are added to the sham group to maximize the null effect. –4.18 (–5.92, –2.44) attacks/month; $p < 0.001$ –4.39 (–6.13, –2.65) attacks/month; $p < 0.001$ Even under this highly pessimistic scenario, the active-vs-sham difference remains statistically significant ($p < 0.001$), confirming the robustness of the primary finding.

Sensitivity analysis excluding patients with baseline attack frequency > 20/month To rule out that extreme baseline values drove the effect, we re-ran the analysis after excluding 4 participants (2 supraorbital, 2 infraorbital) with >20 monthly attacks at baseline. –6.01 (–8.12, –3.90) attacks/month; $p < 0.001$ –6.42 (–8.50, –4.34) attacks/month; $p < 0.001$ Exclusion of extreme baseline values does not change the direction or significance of any comparison.

3.5. Adverse Events and Tolerability

In both active intervention groups, the most commonly reported adverse effects were mild, expected stimulation-related sensations at the electrode site. These were recorded by open-ended clinician inquiry: itching was reported by 9/28 (32.1%) participants in the supraorbital group and 8/27 (29.6%) in the infraorbital group; paresthesia/tingling was reported by 12/28 (42.9%) and 11/27 (40.7%), respectively. All sensations were transient, did not require treatment discontinuation, and were consistent with the known tolerability profile of TENS. No serious adverse events were reported in any group, including the sham arm. The open-ended capture method is acknowledged as a limitation (Section 4.7), as milder or non-obvious effects may have been under-reported.

4. Discussion

4.1. Overview of Main Findings

The present randomized sham-controlled trial was designed to compare the therapeutic effects of TENS applied over two major sensory branches of the trigeminal nerve — the supraorbital and infraorbital nerves — in patients with migraine. Both stimulation protocols resulted in significant improvements in the primary outcome (monthly migraine attack frequency) and in secondary outcomes. At 3 months, both interventions produced comparable clinical benefits relative to sham, with no statistically significant superiority between stimulation sites; because the trial was neither powered nor pre-specified for equivalence or non-inferiority testing, this finding should not be interpreted as proven equivalence, and it should be weighed alongside the site-mismatched sham design described in Section 2.7.

Beyond statistical significance, the magnitude of improvement appears clinically meaningful: reductions of approximately 45–50% in pain intensity, together with substantial improvements in MIDAS and HIT-6 scores, suggest that the observed benefits could translate into tangible improvements in daily functioning and quality of life. However, interpretation of the active-versus-sham contrasts must account for the facts that (a) all outcomes were patient-reported, (b) participants were not fully blinded (Section 2.4), and (c) the randomized sample was pre-selected for high treatment commitment (Section 2.2); together these raise the possibility that expectation effects and a favorably selected cohort contributed to some portion of the observed effect sizes, alongside a genuine physiological effect.

4.2. Comparison with Existing Literature

Our results align with the growing body of evidence supporting neuromodulation as a non-pharmacological intervention for migraine. The TEAM study (enrollment 2018–2019), the largest multicenter randomized trial evaluating external trigeminal nerve stimulation (eTNS) for acute migraine treatment, reported 2-hour pain freedom of 25.5% with active stimulation versus 18.3% with sham ($p = 0.043$), and freedom from the most bothersome symptom of 56.4% versus 42.3% ($p = 0.001$) [25]. Although TEAM assessed acute rather than preventive outcomes, it supports the broader clinical relevance of trigeminal neuromodulation. A systematic review and meta-analysis of neuromodulation RCTs similarly found that selected techniques, including supraorbital TENS for prevention, can improve migraine-related outcomes with a generally favorable safety profile; however, overall evidence quality remains limited and effect sizes are typically small to moderate [26].

Our findings are directionally consistent with this literature. We observed a 46.3% reduction in pain intensity (VAS) in the supraorbital group and 44.8% in the infraorbital group at 3 months, compared with only 8.0% in the sham group (between-group $p < 0.001$ at 3 months). These results are broadly comparable to those of Jiang et al., who reported a $\geq 50\%$ responder rate of 39.22% with transcutaneous supraorbital neurostimulation alone, similar to flunarizine monotherapy (46.15%; $p = 0.90$) [27]; the larger relative reductions observed here should be interpreted cautiously given the between-study differences in design and the selection considerations noted above.

4.3. Site Comparison: Temporal Pattern and Candidate Mechanisms

The numerical difference in the temporal pattern of response between the two active sites was not statistically significant, and the trial was not designed to test it; any interpretation below is therefore hypothesis-generating only and should not be read as evidence of site-specific superiority. With that caveat, supraorbital stimulation showed a numerically faster initial response (44.2% pain reduction at 1 month, 46.3% at 3 months), while infraorbital stimulation showed a more gradual trajectory (28.6% at 1 month, 44.8% at 3 months), converging to a similar level by 3 months (Supra–infra MD 0.02, 95% CI -0.60 to 0.64).

If replicated in adequately powered, site-matched trials, one speculative and untested explanation is anatomical: the supraorbital nerve is a superficial terminal branch of the ophthalmic division of the trigeminal nerve, which may allow faster afferent recruitment, whereas the infraorbital nerve (maxillary division) may engage partially distinct afferent pathways before converging on shared trigeminovascular and brainstem circuits [28,29]. We emphasize that this account is offered only as a possible direction for future mechanistic study, not as an explanation supported by the present data. More broadly, proposed mechanisms for TENS analgesia in migraine include gate-control-mediated suppression of nociceptive transmission via large-diameter A β afferents [13], engagement of descending inhibitory pathways (periaqueductal gray, rostral ventromedial medulla, anterior cingulate cortex, hypothalamus, thalamus) [3,28,29], and modulation of serotonergic, GABAergic, and CGRP-related signaling [29]. The similar 3-month outcomes across sites are compatible with convergence of trigeminal sensory inputs within the trigeminocervical complex, but this trial did not include imaging or biomarker measures capable of testing any of these mechanisms directly.

4.4. Disability, Headache Impact, and Medication Use

Significant improvements were observed in both disability (MIDAS) and headache impact (HIT-6). MIDAS decreased by 51.2% in the supraorbital group and 42.6% in the infraorbital group (both $p < 0.001$). HIT-6 decreased by approximately 20% in both active groups, moving from the severe-impact range (≥ 60) to little or no impact in the supraorbital group (49.48) and to some impact in the infraorbital group (51.40) [32]. A reduction of approximately 5 points on the HIT-6 is generally considered clinically meaningful for within-person change [39]; both active groups exceeded this threshold (supraorbital: -13.1 points; infraorbital: -12.8 points).

Oral analgesic frequency decreased by 65.7% with supraorbital and 61.6% with infraorbital stimulation; MPQ-5 preventive-need scores decreased by 36.5% and 43.9%, respectively. The sham group showed a non-significant 16.0% reduction in analgesic-use frequency ($p = 0.066$) and no significant change in MPQ-5 ($p = 0.507$); absolute analgesic use at 3 months nevertheless remained higher in the sham arm than in either active group. These medication-related findings are clinically relevant given that pharmacological migraine management is often limited by adverse effects, medication-overuse headache, and poor adherence [10,34,35], and are broadly consistent with the concept that peripheral

neuromodulation can, in some settings, reduce medication reliance, although pooled evidence on medication-day outcomes for related modalities such as non-invasive vagus nerve stimulation remains mixed [40].

4.5. Comparison Between Active Groups

The descriptive contrasts below (Table 9) were not statistically significant between the two active arms at 3 months, and the trial was not powered to detect between-site differences; they are presented as hypothesis-generating observations for future, adequately powered, site-matched trials, not as evidence of differential efficacy.

Table 9. Descriptive comparison of active treatment groups. None of the between-site differences below were statistically significant at 3 months; values are presented for hypothesis generation only.

Parameter Supraorbital Infraorbital Interpretation

Onset of pain relief Faster numerically (44.2% at 1 month) Slower numerically (28.6% at 1 month) Descriptive only — not statistically tested

3-month pain reduction 46.3% 44.8% Similar numerically; not tested for equivalence

MIDAS improvement 51.2% 42.6% Numerical trend only

MPQ-5 preventive-need reduction 36.5% 43.9% Numerical trend only

HIT-6 reduction 20.9% (–13.1 pts) 20.0% (–12.8 pts) Similar numerically; not tested for equivalence

4.6. Clinical Implications

Because 3-month outcomes were statistically similar between sites, either stimulation site may be clinically acceptable pending confirmatory research; site selection can reasonably consider electrode practicality, tolerability, and patient preference rather than a claimed efficacy advantage. The non-invasive nature, favorable safety profile (Section 3.5), and demonstrated efficacy against sham support TENS as a candidate option within contemporary migraine management pathways; European expert guidance and American Headache Society consensus support neuromodulation for selected patients, particularly those refractory to or intolerant of pharmacological therapies [36,37]. Because the present protocol evaluated repeated preventive sessions with follow-up to 3 months, claims of acute abortive efficacy would require dedicated acute-treatment trial designs. TENS may also be considered as one component of multidisciplinary migraine rehabilitation programs that include lifestyle modification, physiotherapy, exercise therapy, stress management, and psychological interventions [12,36].

4.7. Methodological Strengths

Strengths include a randomized sham-controlled design, investigator-blinded outcome assessment and analysis, matched session procedures across arms for the supraorbital comparison, multidimensional outcomes (VAS, MIDAS, HIT-6, migraine frequency, analgesic use, and MPQ-5), successful baseline demographic balance for age and sex, and 3-month follow-up. To the best of our knowledge, this trial is among the first randomized controlled studies to directly compare transcutaneous electrical stimulation of the supraorbital and infraorbital trigeminal branches in migraine.

4.8. Limitations and Future Directions

Several limitations should be considered together when interpreting these results:

1. Pre-randomization selection bias: 54 of 137 screened patients were excluded before randomization specifically because they could not commit to the full 24-session schedule. The randomized sample is therefore enriched for high treatment commitment/adherence, which most plausibly explains the 0% post-randomization attrition and may limit generalizability. However, sensitivity analyses under worst-case imputation scenarios (Table 8) confirmed that the primary result remained statistically significant and clinically meaningful even when all excluded patients were assumed to have zero improvement or even deterioration, suggesting that the main finding is robust and unlikely to be fully attributable to this selection bias.
2. Underpowered between-site comparison: The achieved sample size provided only ~64% power for the assumed 3-attack difference, making the non-significant supraorbital-versus-infraorbital comparison inconclusive rather than evidence of equivalence.
3. Sham-site asymmetry: The sham condition replicated only the supraorbital placement; the infraorbital-active versus sham contrast is not site-matched.
4. Incomplete blinding: Blinding assessment did not confirm successful participant blinding; because all outcomes are patient-reported, expectation effects may have inflated effect estimates.
5. No formal multiplicity correction for secondary outcomes in the original analysis; a Bonferroni check is now reported in Section 2.8 and does not change the pattern of results.
6. Confidence intervals are descriptive rather than formally consistent with the rank-based tests.
7. Short follow-up (3 months) does not establish long-term durability.
8. Adverse events were captured by open-ended inquiry rather than a standardized checklist.
9. Single-center design and predominantly female sample (75.9%) limit generalizability.
10. No subgroup analyses by migraine subtype (episodic vs chronic; with vs without aura).
11. Ethics and IRCT numbers are placeholders and must be completed before submission.

Future directions: Future research should include larger, rigorously blinded, site-matched sham-controlled trials with a pre-specified statistical analysis plan; longer follow-up; mechanistic imaging studies; dose-finding designs; and identification of predictors of treatment response.

5. Conclusions

In this single-blind, sham-controlled randomized trial in a highly adherent population, preventive TENS applied over either the supraorbital or the infraorbital trigeminal branch produced significant within-group improvements in monthly migraine attack frequency (primary outcome), pain intensity (VAS), disability (MIDAS), headache impact (HIT-6), oral analgesic use, and MPQ-5 preventive-need scores. At 3 months, both active protocols were superior to sham across these outcomes, whereas sham showed no significant within-group change. No statistically significant difference was detected between the two active sites at 3 months; given the trial's limited power for this comparison and the sham-site asymmetry, this should be read as inconclusive rather than as evidence of equivalence.

Descriptively, supraorbital stimulation was associated with a faster early reduction in pain intensity, whereas infraorbital stimulation showed a more gradual VAS trajectory and a larger relative reduction in MPQ-5 scores. These temporal patterns are hypothesis-generating only and should not be interpreted as evidence of site-specific superiority.

Taken together, TENS targeting either trigeminal branch may be considered a candidate non-pharmacological adjunctive option for migraine prevention in similar clinical settings, in patients able to commit to the intensive session schedule used here. Confirmation in larger, site-matched, more rigorously blinded trials with longer follow-up, a pre-specified analysis plan, and verifiable prospective registration is warranted before stronger practice recommendations can be made.

Acknowledgments

The authors thank all patients who participated in this study and the clinical staff of the Physical Medicine and Rehabilitation clinics at Isfahan University of Medical Sciences for their assistance with recruitment and follow-up assessments. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics approval and consent to participate

The study protocol was approved by the institutional ethics committee of Isfahan University of Medical Sciences (ethics approval code: []). All participants provided written informed consent. The trial was prospectively registered in the Iranian Registry of Clinical Trials (IRCT registration number: []). The study was conducted in accordance with the Declaration of Helsinki.

Competing interests

The authors declare that they have no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

De-identified individual participant data and the statistical analysis code underlying the results of this article are available from the corresponding author upon reasonable request, subject to institutional ethics requirements.

Authors' contributions

All authors contributed to study conception and design, data acquisition or interpretation, and critical revision of the manuscript. All authors read and approved the final manuscript. Detailed CRediT roles may be finalized by the authors before submission.

References

1. GBD 2016 Headache Collaborators. Global, regional, and national burden of migraine and tension-type headache, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol.* 2018;17(11):954–976. doi:10.1016/S1474-4422(18)30322-3
2. Headache Classification Committee of the International Headache Society (IHS). The International Classification of Headache Disorders, 3rd edition. *Cephalalgia.* 2018;38(1):1–211. doi:10.1177/0333102417738202
3. Akerman S, Romero-Reyes M, Holland PR. Current and novel insights into the neurophysiology of migraine and its implications for therapeutics. *Pharmacol Ther.* 2017;172:151–170. doi:10.1016/j.pharmthera.2016.12.005
4. Tepper SJ, Dahlöf CG, Dowson A, Newman L, Mansbach H, Jones M, Pham B, Webster C, Salonen R. Prevalence and diagnosis of migraine in patients consulting their physician with a complaint of headache: data from the Landmark Study. *Headache.* 2004;44(9):856–864. doi:10.1111/j.1526-4610.2004.04167.x

5. Weatherall MW. Acute neurology in a twenty-first century district general hospital. *J R Coll Physicians Edinb.* 2006;36(3):196–200.
6. Stone J, Carson A, Duncan R, Roberts R, Warlow C, Hibberd C, Coleman R, Cull R, Murray G, Pelosi A, Cavanagh J, Matthews K, Goldbeck R, Smyth R, Walker J, Sharpe M. Who is referred to neurology clinics?—the diagnoses made in 3781 new patients. *Clin Neurol Neurosurg.* 2010;112(9):747–751. doi:10.1016/j.clineuro.2010.05.011
7. Welch KMA, Goadsby PJ. Chronic daily headache: nosology and pathophysiology. *Curr Opin Neurol.* 2002;15(3):287–295.
8. Natoli JL, Manack A, Dean B, Butler Q, Turkel CC, Stovner L, Lipton RB. Global prevalence of chronic migraine: a systematic review. *Cephalalgia.* 2010;30(5):599–609. doi:10.1111/j.1468-2982.2009.01941.x
9. Buse D, Manack A, Serrano D, Reed M, Varon S, Turkel C, Lipton R. Headache impact of chronic and episodic migraine: results from the American Migraine Prevalence and Prevention study. *Headache.* 2012;52(1):3–17. doi:10.1111/j.1526-4610.2011.02046.x
10. Kristoffersen ES, Lundqvist C. Medication-overuse headache: epidemiology, diagnosis and treatment. *Ther Adv Drug Saf.* 2014;5(2):87–99. doi:10.1177/2042098614522683
11. Magis D, Jensen R, Schoenen J. Neurostimulation therapies for primary headache disorders: present and future. *Curr Opin Neurol.* 2012;25(3):269–278. doi:10.1097/WCO.0b013e3283532023
12. Ambrosini A, D'Alessio C, Magis D, Schoenen J. Targeting pericranial nerve branches to treat migraine: current approaches and perspectives. *Cephalalgia.* 2015;35(14):1308–1322. doi:10.1177/0333102415573511
13. Johnson MI, Paley CA, Howe TE, Sluka KA. Transcutaneous electrical nerve stimulation for acute pain. *Cochrane Database Syst Rev.* 2015;(6):CD006142. doi:10.1002/14651858.CD006142.pub3
14. Tao H, Wang T, Dong X, Guo Q, Xu H, Wan Q. Effectiveness of transcutaneous electrical nerve stimulation for the treatment of migraine: a meta-analysis of randomized controlled trials. *J Headache Pain.* 2018;19(1):42. doi:10.1186/s10194-018-0868-9
15. Schoenen J, Vandersmissen B, Jeanette S, Herroelen L, Vandenheede M, Gérard P, Magis D. Migraine prevention with a supraorbital transcutaneous stimulator: a randomized controlled trial. *Neurology.* 2013;80(8):697–704. doi:10.1212/WNL.0b013e3182825055
16. Magis D, Sava S, d'Elia TS, Baschi R, Schoenen J. Safety and patients' satisfaction of transcutaneous supraorbital neurostimulation (tSNS) with the Cefaly® device in headache treatment: a survey of 2,313 headache sufferers in the general population. *J Headache Pain.* 2013;14:95. doi:10.1186/1129-2377-14-95
17. Gérardy PY, Fabry D, Fumal A, Schoenen J. A pilot study on supra-orbital surface electrotherapy in migraine. *Cephalalgia.* 2009;29:134.
18. Russo A, Tessitore A, Conte F, Marcuccio L, Giordano A, Tedeschi G. Transcutaneous supraorbital neurostimulation in "de novo" patients with migraine without aura: the first Italian experience. *J Headache Pain.* 2015;16:69. doi:10.1186/s10194-015-0551-3
19. Vikelis M, Dermitzakis EV, Spingos KC, Vasiliadis GG, Vlachos GS, Kararizou E. Clinical experience with transcutaneous supraorbital nerve stimulation in patients with refractory migraine or with migraine and intolerance to topiramate: a prospective exploratory clinical study. *BMC Neurol.* 2017;17:97. doi:10.1186/s12883-017-0869-3

20. Birlea M, Penning S, Callahan K, Schoenen J. Efficacy and safety of external trigeminal neurostimulation in the prevention of chronic migraine: an open-label trial. *Cephalalgia Rep.* 2019;2:1–10. doi:10.1177/2515816319856625
21. Tajti J, Szok D, Nyári A, Vécsei L. Therapeutic strategies that act on the peripheral nervous system in primary headache disorders. *Expert Rev Neurother.* 2019;19(6):509–528.
22. Liu Y, Dong Z, Wang R, Ao R, Han X, Tang W, Yu S. Migraine prevention using different frequencies of transcutaneous occipital nerve stimulation: a randomized controlled trial. *J Pain.* 2017;18(8):1006–1015. doi:10.1016/j.jpain.2017.03.012
23. Ilhan Alp S, Alp R. Supraorbital and infraorbital nerve blockade in migraine patients: results of 6-month clinical follow-up. *Eur Rev Med Pharmacol Sci.* 2013;17(13):1778–1781.
24. Reed KL, Conidi F, Bulger R, Will K. Combined concordant peripheral neurostimulation for chronic migraine headaches: a retrospective analysis of 188 consecutive patients. *Neurology.* 2014;82(10 Suppl):S41.001.
25. Kuruvilla DE, Mann JI, Tepper SJ, Starling AJ, Panza J, Johnson MA, Mullin C, et al. Phase 3 randomized, double-blind, sham-controlled Trial of e-TNS for the Acute treatment of Migraine (TEAM). *Sci Rep.* 2022;12:5110. doi:10.1038/s41598-022-09071-6
26. Moisset X, Pereira B, Ciampi de Andrade D, Fontaine D, Lantéri-Minet M, Mawet J. Neuromodulation techniques for acute and preventive migraine treatment: a systematic review and meta-analysis of randomized controlled trials. *J Headache Pain.* 2020;21(1):142. doi:10.1186/s10194-020-01204-4
27. Jiang L, Yuan DL, Li M, Liu C, Liu Q, Zhang Y, Tan G. Combination of flunarizine and transcutaneous supraorbital neurostimulation improves migraine prophylaxis. *Acta Neurol Scand.* 2019;139(3):276–283. doi:10.1111/ane.13050
28. Goadsby PJ, Holland PR, Martins-Oliveira M, Hoffmann J, Schankin C, Akerman S. Pathophysiology of migraine: a disorder of sensory processing. *Physiol Rev.* 2017;97(2):553–622. doi:10.1152/physrev.00034.2015
29. Powell K, Lin K, Tambo W, Saavedra AP, Sciubba D, Al Abed Y, Li C. Trigeminal nerve stimulation: a current state-of-the-art review. *Bioelectron Med.* 2023;9(1):30. doi:10.1186/s42234-023-00128-z
30. Westwood SJ, Conti AA, Tang W, Xue S, Cortese S, Rubia K. Clinical and cognitive effects of external trigeminal nerve stimulation (eTNS) in neurological and psychiatric disorders: a systematic review and meta-analysis. *Mol Psychiatry.* 2023;28(10):4025–4043. doi:10.1038/s41380-023-02227-4
31. Lipton RB, Serrano D, Buse D, Bigal M, Rupnow M, Biondi D. The Migraine Prevention Questionnaire (MPQ-5): development, psychometric testing and validation. *Headache.* 2007;47(5):770–771.
32. Yang M, Rendas-Baum R, Varon SF, Kosinski M. Validation of the Headache Impact Test (HIT-6™) across episodic and chronic migraine. *Cephalalgia.* 2011;31(3):357–367. doi:10.1177/0333102410379890
33. Stewart WF, Lipton RB, Kolodner K, Liberman J, Sawyer J. Reliability of the migraine disability assessment score in a population-based sample of headache sufferers. *Cephalalgia.* 1999;19(2):107–114. doi:10.1046/j.1468-2982.1999.019002107.x
34. Steiner TJ, Stovner LJ, Jensen R, Uluduz D, Katsarava Z; Lifting The Burden: the Global Campaign against Headache. Migraine remains second among the world's causes of disability,

and first among young women: findings from GBD2019. *J Headache Pain*. 2020;21(1):137. doi:10.1186/s10194-020-01208-0

35. Diener HC, Holle D, Solbach K, Gaul C. Medication-overuse headache: risk factors, pathophysiology and management. *Nat Rev Neurol*. 2016;12(10):575–583. doi:10.1038/nrneurol.2016.124

36. Martelletti P, Jensen RH, Antal A, Arcioni R, Brighina F, de Tommaso M, Franzini A, Fontaine D, Heiland M, Jürgens TP, Leone M, Magis D, Paemeleire K, Palmisani S, Paulus W, May A. Neuromodulation of chronic headaches: position statement from the European Headache Federation. *J Headache Pain*. 2013;14:86. doi:10.1186/1129-2377-14-86

37. Ailani J, Burch RC, Robbins MS; Board of Directors of the American Headache Society. The American Headache Society Consensus Statement: Update on integrating new migraine treatments into clinical practice. *Headache*. 2021;61(7):1021–1039. doi:10.1111/head.14153

38. Zandifar A, Asgari F, Haghdooost F, Masjedi SS, Manouchehri N, Banihashemi M, et al. Reliability and validity of the Persian HIT-6 questionnaire in migraine and tension-type headache. *Pain Pract*. 2014;14(5):e161–e166. doi:10.1111/papr.12163

39. Smelt AFH, Assendelft WJJ, Terwee CB, Ferrari MD, Blom JW. What is a clinically relevant change on the HIT-6 questionnaire? An estimation in a primary-care population of migraine patients. *Cephalalgia*. 2014;34(1):29–36. doi:10.1177/0333102413497599

40. Song D, Li P, Wang Y, Cao J. Noninvasive vagus nerve stimulation for migraine: a systematic review and meta-analysis of randomized controlled trials. *Front Neurol*. 2023;14:1190062. doi:10.3389/fneur.2023.1190062

Tab 2

